

Problem 1

```
[[ 2  1 -1]
 [ 3  4  2]
 [ 1 -2  5]]
[[3]
 [7]
 [4]]
```

Part a

```
[[1.66666667]
 [0.22222222]
 [0.55555556]]
```

Part b

45.000000000000014

Part c

```
[[ 2  3  1]
 [ 1  4 -2]
 [-1  2  5]]
```

Part d

11

Part e

3

Part f

3 = 3 ? True

Part g

```
[[14 12  9]
 [12 21 -3]
 [ 9 -3 30]]
```

False, because $A^T * A$ is not equal to identity matrix

Part h

A Inverse:

```
[[ 0.53333333 -0.06666667  0.13333333]
 [-0.28888889  0.24444444 -0.15555556]
 [-0.22222222  0.11111111  0.11111111]]
```

A Transpose

```
[[ 2  3  1]
 [ 1  4 -2]
 [-1  2  5]]
```

False, A Inverse is not equal to A Transpose

Part i

Eigenvalues: [2.13629347+0.j 4.43185326+1.19297873j 4.43185326-1.19297873j]

Eigenvectors: [[-0.67303792+0.j -0.32467593-0.04622835j -0.32467593+0.04622835j]

[0.47510141+0.j 0.06465707-0.49975204j 0.06465707+0.49975204j]

[0.56683208+0.j 0.79907186+0.j 0.79907186-0.j]]

Part j

Graident:

Matrix([[2*x + 3*y + 2*z, 3*x + 2*y, 2*x + 2*z]])

Hessian:

Matrix([[2, 3, 2], [3, 2, 0], [2, 0, 2]])

Problem 2

```
[[ 14 -7 -10]
 [  2  9  5]
 [ 14 -22 16]]
[[2]
 [8]
 [6]]
[[13]
 [21]
 [16]]
```

Part a

```
[[290]]
```

Part b

```
[[ -88]
 [106]
 [-52]]
```

Part c

```
[[ 396 -388  94]
 [-388  614 -237]
 [  94 -237 381]]
```

Part d

False because M transposed is not equal to M

Part e

```
v1 L2 Norm: 10.19803902718557
v1 L1 Norm: 16.0
v1 LInfinity Norm: 8.0
v2 L2 Norm: 29.42787793912432
v2 L1 Norm: 50.0
v2 LInfinity Norm: 21.0
```

Problem 3

	Transaction ID	Item	Quantity	Price	Per Unit	Total	Spent	Payment Method	Location	Transaction Date
0	TXN_1961373	Coffee	2		2.0		4.0	Credit Card	Takeaway	2023-09-08
1	TXN_4977031	Cake	4		3.0		12.0	Cash	In-store	2023-05-16
2	TXN_4271903	Cookie	4		1.0		ERROR	Credit Card	In-store	2023-07-19
3	TXN_7034554	Salad	2		5.0		10.0	UNKNOWN	UNKNOWN	2023-04-27
4	TXN_3160411	Coffee	2		2.0		4.0	Digital Wallet	In-store	2023-06-11

Chosen Column: Total Spent

Part a

	Before	After
Mean	8.924352	8.936711
Variance	36.119132	36.392933
Standard Deviation	6.009919	6.032656
# Rows	9498.000000	3089.000000

Part b

	Before	After
Mean	8.924352	8.924352
Variance	36.119132	34.305770
Standard Deviation	6.009919	5.857113
# Rows	9498.000000	10000.000000

Part c

	Before	After
Mean	8.924352	8.877950
Variance	36.119132	34.346513
Standard Deviation	6.009919	5.860590
# Rows	9498.000000	10000.000000

Part d

	Before	After
Mean	8.924352	8.911950
Variance	36.119132	35.966819
Standard Deviation	6.009919	5.997234
# Rows	9498.000000	10000.000000

Part e

	Before	After
Mean	8.924352	8.932900
Variance	36.119132	36.152713
Standard Deviation	6.009919	6.012713
# Rows	9498.000000	10000.000000

Part f

	Before	After
Mean	8.924352	8.913808
Variance	36.119132	36.027922
Standard Deviation	6.009919	6.002326
# Rows	9498.000000	9357.000000